Australian Mathematics Trust

Australian Mathematics Competition

AN ACTIVITY OF THE AUSTRALIAN MATHEMATICS TRUST

NAME _____

YEAR _____ TEACHER _____

2018
CHINESE VERSION
UPPER PRIMARY DIVISION

AUSTRALIAN SCHOOL YEARS 5, 6 and PRIMARY 7*

*SOME STATES ONLY

TIME ALLOWED: 60 MINUTES

INSTRUCTIONS AND INFORMATION

GENERAL

1. Do not open the booklet until told to do so by your teacher.
2. You may use any teaching aids normally available in your classroom, such as MAB blocks, counters, currency, calculators, play money etc. You are allowed to work on scrap paper and teachers may explain the meaning of words in the paper. Mobile phones are not permitted.
3. Diagrams are NOT drawn to scale. They are intended only as aids.
4. There are 25 multiple-choice questions, each requiring a single answer, and 5 questions that require a whole number answer between 0 and 999. The questions generally get harder as you work through the paper. There is no penalty for an incorrect response.
5. This is a competition not a test; do not expect to answer all questions. You are only competing against your own year in your own country/Australian state so different years doing the same paper are not compared.
6. Read the instructions on the answer sheet carefully. Ensure your name, school name and school year are entered. It is your responsibility to correctly code your answer sheet.
7. When your teacher gives the signal, begin working on the problems.

THE ANSWER SHEET

1. Use only lead pencil.
2. Record your answers on the reverse of the answer sheet (not on the question paper) by FULLY colouring the circle matching your answer.
3. Your answer sheet will be scanned. The optical scanner will attempt to read all markings even if they are in the wrong places, so please be careful not to doodle or write anything extra on the answer sheet. If you want to change an answer or remove any marks, use a plastic eraser and be sure to remove all marks and smudges.

INTEGRITY OF THE COMPETITION

The AMT reserves the right to re-examine students before deciding whether to grant official status to their score.

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1-10 题，每题 3 分
Questions 1 to 10, 3 marks each

1. 请问下列哪一项内的数最接近 208?

Which one of these numbers is closest to 208?

(A) 190 (B) 200 (C) 205 (D) 210 (E) 218

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2. 小佳原本有 \$47，她又得到 \$25 的生日礼金。请问她现在有多少钱?

Callie has \$47 and then gets \$25 for her birthday. How much does she have now?

(A) \$52 (B) \$62 (C) \$65 (D) \$69 (E) \$72

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3. 请问下列哪一项内的数是 8 的倍数?

Which one of the following numbers is a multiple of 8?

(A) 18 (B) 28 (C) 38 (D) 48 (E) 58

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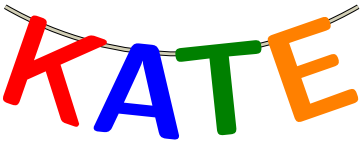
4. 小凯用字母串珠做出右侧这个项链。她将它翻面过来，展现出串珠的背面。请问下列哪一项内的图是现在看起来的样子?

Kate made this necklace from alphabet beads. She put it on the wrong way around, showing the back of the beads. What does this look like?



(A)

(B)

(C)

(D)

(E)

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5. 请写出八仟八佰零八的阿拉伯数。

Write the number for eight thousand, eight hundred and eight.

(A) 88008 (B) 80808 (C) 80088 (D) 888 (E) 8808

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6. 小珍有一些 20 分的硬币而小德有一些 50 分的硬币。若他们两人有一样多的钱。请问他们两人的硬币总数量最少有多少枚?

Jane has a number of 20c coins and Tariq has a number of 50c coins. They have the same amount of money. What is the smallest number of coins they could have all together?

(A) 2 (B) 5 (C) 6 (D) 7 (E) 10

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7. 张太太将 58 本书放回图书馆的书架上。除了最后一个书架外，她在每一个书架上都放了 12 本书。请问她在最后一个书架上放置几本书?

Mrs Chapman put 58 books back on the library shelves. She put 12 books on each shelf except the last shelf. How many books did she put on the last shelf?

(A) 7 (B) 8 (C) 9 (D) 10 (E) 11

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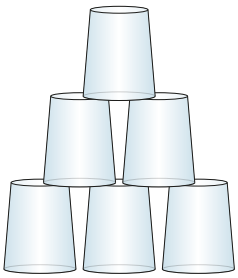
8. 有一位店员将塑料杯依右图所示方式摆放，每一层都比底下一层少一个杯子，而最顶层只有一个杯子。
她保持此型式摆放，直到总共摆放 28 个杯子为止。请问杯子共有多少层?



A shopkeeper displays plastic cups like this. Each level has one less than the level below it, and the top level has only one cup.
She keeps this pattern going until she has 28 cups. How many levels is this?

(A) 5 (B) 6 (C) 7 (D) 8 (E) 9

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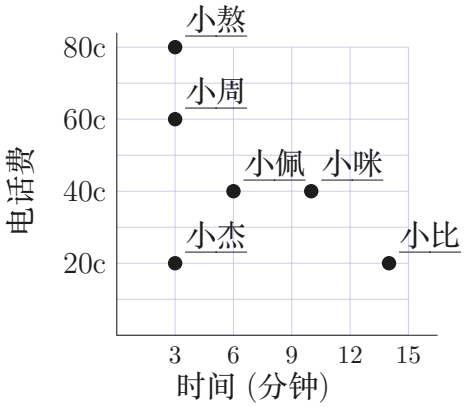
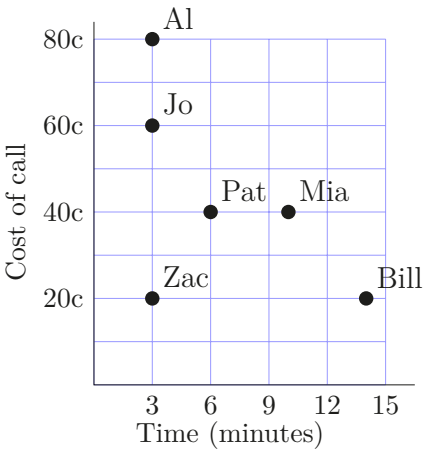
9. 六个朋友分别打了一通电话到其它的城市。电话费是依照打电话的时间长短与距离计价。
从右图中，请问哪一个人打电话的时间比小佩长但电话费却较少?

(A) 小熬 (B) 小比 (C) 小周 (D) 小咪 (E) 小杰



Six friends each make a phone call to another city.
The cost of each call depends on the time taken for the call as well as the distance.
From this diagram decide whose phone call lasts longer than Pat's, but costs less.

(A) Al (B) Bill (C) Jo (D) Mia (E) Zac



10. A、B、C 三人比较他们之间的年龄。A 现年 8 岁，再过三年，B 将是 9 岁。在二年前，C 是 5 岁。
依照由年龄小到大排序，其顺序为

(A) A、B、C (B) B、C、A (C) C、A、B (D) B、A、C (E) A、C、B

Aimee, Bilal and Caitlin are comparing their ages. Aimee is 8 years old. In three years time, Bilal will be 9. Two years ago, Caitlin was 5.
Ordered from youngest to oldest, they are

(A) Aimee, Bilal, Caitlin (B) Bilal, Caitlin, Aimee (C) Caitlin, Aimee, Bilal (D) Bilal, Aimee, Caitlin (E) Aimee, Caitlin, Bilal

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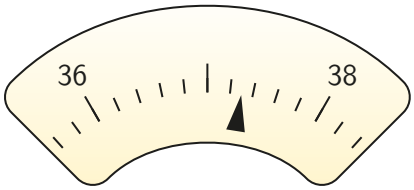
11-20 题，每题 4 分
Questions 11 to 20, 4 marks each

11. 请问这个普遍使用的仪表所指出的值是什么？

What value is indicated on this popularity meter?



(A) 36.65 (B) 37.15 (C) 37.3 (D) 37.65 (E) 38.65

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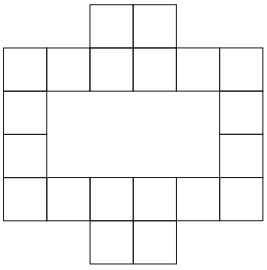
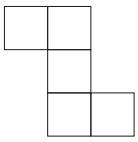
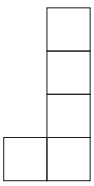
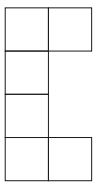
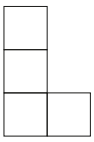
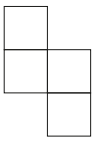
12. 用选项内的多方块之一若干片可以构成右侧的图案，多方块可以旋转、翻转，但不可以重迭。请问是哪一个选项？

One of these shapes made of squares has been flipped and turned to make the following pattern, without any overlaps. Which one?



(A)  (B)  (C)  (D)  (E) 

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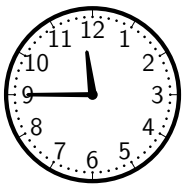
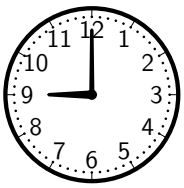
13. 小飞在星期五的课表如图所示，当在上图书馆课时他看着时钟。
请问此时可能是下列哪一项内时钟所指的时刻?

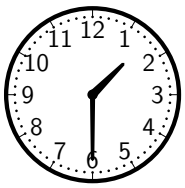
星期五课表	
9:00 am	英文
10:00 am	数学
11:00 am	休息
11:30 am	图书馆
12:30 pm	集会
1:00 pm	午餐
2:00 pm	体育

Fred looked at the clock during the Library lesson.
Which one of these times could the clock have shown?

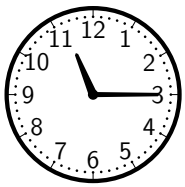
Friday timetable	
9.00 am	English
10.00 am	Mathematics
11.00 am	Recess
11.30 am	Library
12.30 pm	Assembly
1.00 pm	Lunch
2.00 pm	Sport

(A) 

(B) 

(C) 

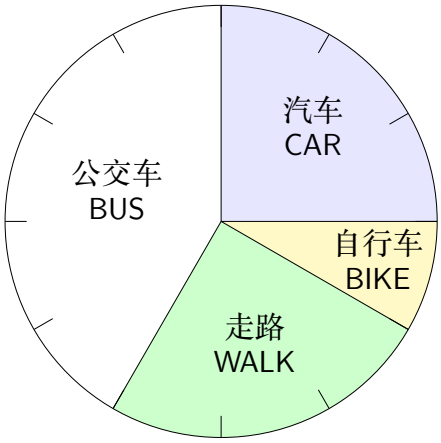
(D)

(E) 

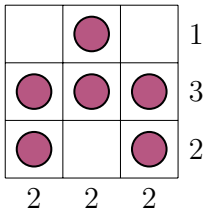
14. 小亚在去年工作 48 周，每周 5 天。右表指出小亚每天上班的方式。请问小亚总共有几天搭公交车上班?

Last year Alan worked 5 days a week for 48 weeks. The graph shows how Alan travelled to work each day. On how many days did Alan travel by bus?

(A) 20 (B) 80 (C) 100 (D) 140 (E) 240

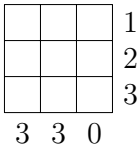


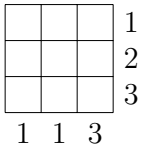
15. 如右图所示的方格表中，在一行的末端或一列的底端上的每个数是指在这一行或这一列的方格内包括有多少个圆圈。请问下列哪一个方格表内可能有圆圈也符合此规则？

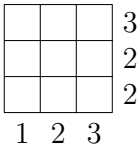


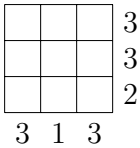
In this grid, each number at the end of a row or below a column indicates how many squares in that row or column contain a counter.

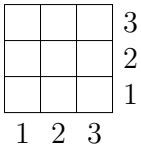
Which one of the following grids could also have counters with these rules?

(A) 

(B) 

(C) 

(D) 

(E) 

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16. 邮寄大件国际包裹，首 10 kg 邮资为 \$24，超过者每续 5 kg(不足 5 kg 以 5 kg 计)加收 \$8。请问邮寄一件 28 kg 的国际包裹需要多少邮资？

To send large parcels overseas, it costs \$24 for the first 10 kg and \$8 for each extra 5 kg or part thereof. How much would it cost to send a 28 kg parcel overseas?

(A) \$48 (B) \$52 (C) \$56 (D) \$60 (E) \$64

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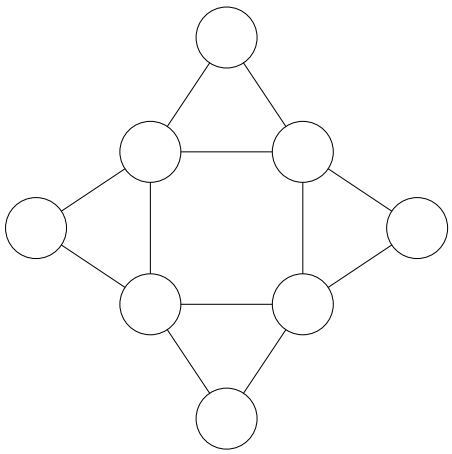
17. 将从 1 到 3 的数填入如图所示的圆圈内。由同一条线相连的两个圆圈内不可以填入相同的数。有许多种填入的方法。请问所填入的八个数之最小可能总和是多少？



The numbers from 1 to 3 are entered into the circles in the grid shown. Two circles joined by a line may not contain the same number.

There are several ways of doing this. What is the smallest possible total of the eight numbers?

(A) 10 (B) 12 (C) 14 (D) 15 (E) 16

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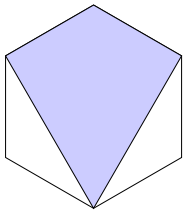
18. 请问这个正六边形的几分之几被涂上阴影？

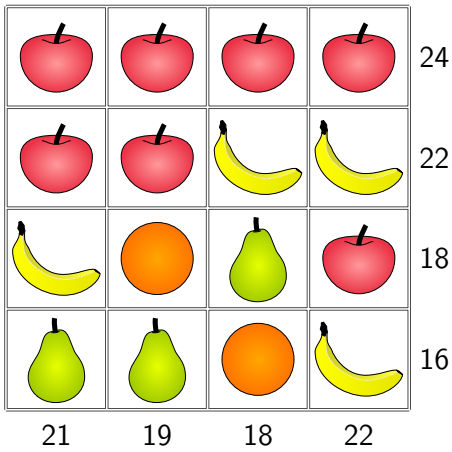


What fraction of this regular hexagon is shaded?

(A) $\frac{1}{2}$ (B) $\frac{2}{3}$ (C) $\frac{3}{4}$ (D) $\frac{3}{5}$ (E) $\frac{4}{5}$

19. 一些水果的图片放在这个方格表内，它们代表小于 10 的正整数。
每一列与每一行的总和已经分别列出。
请问一颗苹果 与一颗橘子 的总和是多少?

Pictures of fruit have been placed in this grid to represent numbers less than 10.
The totals for each row and column are shown.
What is the total value of an apple and an orange ?



(A) 8 (B) 9 (C) 10 (D) 11 (E) 12

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20. 小安打算进行一些打扫工作。他打扫 2 间大房间的时间与打扫 3 间小房间所用的时间相同。他可以用 90 分钟打扫一间大房间与三间小房间。
请问他打扫 3 间大房间与 6 间小房间需费时多少时间?

(A) 3.5 小时 (B) 4 小时 (C) 4.5 小时 (D) 5 小时 (E) 5.5 小时

Andrew is doing some tidying. He can tidy 2 big rooms in the same time it takes to tidy 3 small rooms. He can tidy one big room and three small rooms in 90 minutes.
How long will it take him to tidy 3 big rooms and 6 small rooms?

(A) 3.5 hours (B) 4 hours (C) 4.5 hours (D) 5 hours (E) 5.5 hours

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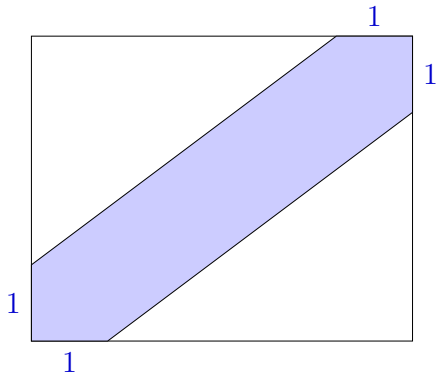
21-25 题，每题 5 分
Questions 21 to 25, 5 marks each

21. 有一个 3 cm × 4 cm 的矩形。从距两个对角顶点各 1 cm 处开始的一条斜对角带子涂上阴影，如图所示。
请问这条涂上阴影的带子之面积占整个矩形面积的几分之几?

A rectangle measures 3 cm by 4 cm. A diagonal stripe is shaded which starts 1 cm from the diagonal corners, as in the diagram.
What fraction of the area of the rectangle is this shaded strip?



(A) $\frac{1}{2}$ (B) $\frac{1}{3}$ (C) $\frac{1}{4}$ (D) $\frac{1}{5}$ (E) $\frac{2}{5}$

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22. 初始时 20 枚硬币排列成一行，A 先取走第一枚硬币，接着再将之后的每第四枚硬币取走。
接着从剩余的硬币中，B 取走第一枚硬币，接着再将之后的每第三枚硬币取走。
接着从剩余的硬币中，C 取走第一枚硬币，接着再将之后的每第二枚硬币取走。
最后 D 取走所有剩下的硬币。
请问是否有谁比其它人取走的硬币都来得多？

(A) 是的，A 取得最多　(B) 是的，B 取得最多　(C) 是的，C 取得最多
(D) 是的，D 取得最多　(E) 没有，他们取得硬币的数量一样多

Beginning with a row of 20 coins, Anh takes the first coin, then every fourth coin after that.
From the remaining coins, Brenda takes the first coin and every third coin after that.
From the remaining coins, Chen takes the first coin and every second coin after that.
Dimitris takes all the remaining coins.
Does anyone get more coins than all the others?

(A) Yes, Anh does　(B) Yes, Brenda does　(C) Yes, Chen does
(D) Yes, Dimitris does　(E) No, they all get the same number of coins

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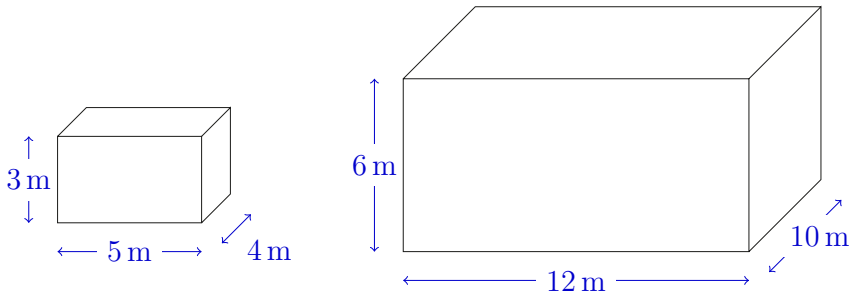
23. 这二个水槽将被注满水。有一条水管可以用 2 小时将小水槽注满。请问用同一条水管将大水槽注满需多少小时？



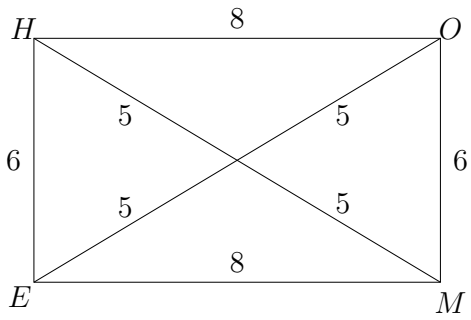
These two water tanks are to be filled. A hose used to do this can fill the smaller tank in 2 hours. How many hours will the same hose take to fill the larger tank?

(A) 4　(B) 6　(C) 9　(D) 12　(E) 24

24. 一位农夫有一块 8 km ×6 km 的矩形土地。在土地四周及对角线上用篱笆围起来，如图所示。

某一天这位农夫离开位于 H 点的家巡视所有的篱笆，当完成后最后再回到家里。

请问这位农夫最少要行走多少 km 才能完成此任务?



A farmer has a rectangular property 8 km by 6 km, with fencing along the boundary and diagonal fences as shown.

One day she leaves her farmhouse at H to inspect all her fences, returning home to H when this is done.

What is the minimum distance, in kilometres, she must travel to do this?

(A) 48 (B) 58 (C) 59 (D) 60 (E) 64

25. 请问以下减式 (其中被减数是 20 个数码都是 1、减数是 10 数码都是 2) 所得的结果之数码和是多少?

$$\underbrace{111\ldots111}_{20\text{ 位数}} - \underbrace{222\ldots222}_{10\text{ 位数}}$$

What is the sum of the digits in the result of the subtraction

$$\underbrace{111\ldots111}_{20\text{ digits}} - \underbrace{222\ldots222}_{10\text{ digits}}$$

where the first number has 20 digits each 1, and the second has 10 digits, each 2?

(A) 72 (B) 81 (C) 89 (D) 90 (E) 99

问题 26-30 的答案为 000-999 之间的整数，
请将答案填在答案卡上对应的位置。

第 26 题占 6 分，第 27 题占 7 分，第 28 题占 8 分，
第 29 题占 9 分，第 30 题占 10 分。

For questions 26 to 30, shade the answer as an integer from 0 to 999 in the space provided on the answer sheet.

Questions 26–30 are worth 6, 7, 8, 9 and 10 marks, respectively.

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26. 在下列的算式中，字母 a、b、c 分别代表 0 到 9 的不同数码。
请问三位数 $\overline{abc}$ 是什么？

In the algorithm below, the letters a, b and c represent different digits from 0 to 9. What is the three-digit number abc?

$$\begin{array}{rcccc} & & & & a \\ & & & a & b \\ & & a & b & c \\ + & 1 & 0 & 0 & 0 \\ \hline & 2 & 0 & 1 & 8 \end{array}$$

---

27. 只使用数码 0、1、2，右图这个正立方体每个面上的数都互不相同。
任意在相对面上的两个数之和都等于一个相同的三位数。
请问这两个数之和的最大可能值是什么？

Using only digits 0, 1 and 2, this cube has a different number on each face.
Numbers on each pair of opposite faces add to the same 3-digit total.
What is the largest that this total could be?

---

28. 我将正整数接续写在一起：

$$1234567891011121314151617\ldots$$

请问当我写出第 100 个零时，我所写的这个数是什么？

I wrote the counting numbers joined together:

$$1234567891011121314151617\ldots$$

Which of the counting numbers was I writing when the 100th zero was written?

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29. 小贞与小吉都在同一个圆形跑道的同一点上。
小吉以匀速跑一圈费时 72 秒。
小贞朝相反方向以匀速行走，每 56 秒与小吉相遇一次。
请问小贞每走一圈费时多少秒?

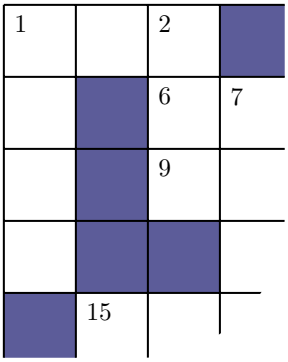
Jan and Jill are both on a circular track.
Jill runs at a steady pace, completing each circuit in 72 seconds.
Jan walks at a steady pace in the opposite direction and meets Jill every 56 seconds.
How long does it take Jan to walk each circuit?

---

30. 解答一个填数益智游戏的提示是一个正整数 (不是一个单字)
如图所示是一个这样的游戏之一小部分。一些提示如下:

横行
1. 是某个数的平方。
6. 等于 1-横行的一半。

直下
1. 两倍的 2-直下。
2. 是个 9 的倍数。

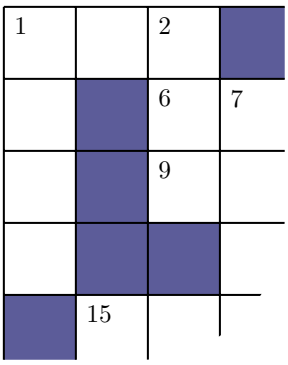


请问 2-直下的数是什么?

The answer to a cross-number puzzle clue is a whole number (not a word).
A fragment of such a puzzle is shown. Some clues are:

Across
1. Square of 27-down.
6. Half of 1-across.

Down
1. Twice 2-down.
2. A multiple of 9.



What is 2-down?

Answer Key

QUES	MP	UP	J	I	S
1	C	D	C	D	D
2	E	E	E	E	C
3	A	D	B	D	D
4	B	A	A	D	B
5	D	E	E	E	A
6	D	D	E	B	A
7	A	D	E	C	C
8	E	C	C	B	C
9	D	B	E	A	E
10	A	B	C	B	A
11	D	C	C	C	D
12	B	A	B	A	E
13	E	A	C	D	E
14	B	C	B	E	E
15	A	E	B	A	C
16	C	C	D	E	B
17	B	C	B	E	B
18	A	B	D	C	A
19	C	B	A	E	D
20	C	A	A	A	C
21	D	A	D	A	C
22	B	E	D	D	A
23	D	E	A	C	C
24	E	D	C	D	B
25	D	D	D	B	E
26	16	918	321	156	90
27	561	321	156	40	97
28	936	509	40	435	90
29	918	252	529	505	118
30	509	693	90	118	101